14 Expected Values of Continuous Random Variables

Example 14.1. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Explain how you could approximate $\mathrm{E}(X)$ as a sum (as if X were a discrete random variable). Hint: recall Example 12.2.

3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

4. Compute $\mathrm{E}(X)$ as an integral. Compare to the approximations.

5. Interpret $\mathrm{E}(X) = 40$ in context.

6. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a *continuous* random variable with pdf f_X is defined as

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average, regardless of whether the variable is discrete or continuous; just sum all the simulated values and divide by the number of simulated values.

Example 14.2. Continuing Example 14.1 where X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

Now we want to compute variance and standard deviation.

1. What is the formula for computing variance?

2. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

3. Explain how you could approximate $\mathrm{E}(X^2)$ as a sum (like a discrete random variable).

4. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

5. Compute $\mathrm{E}(X^2)$ as an integral. (You should get 1800.)

6. Compute $\mathrm{Var}(X)$.

7. Compute and interpret $\mathrm{SD}(X)$.

- The **"law of the unconscious statistician" (LOTUS)** says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \qquad \mathrm{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\text{Continuous } X \text{ with pdf } f_X: \qquad \mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

- LOTUS says we don't have to first find the distribution of $Y = g(X)$ to find $\mathrm{E}[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X)$$

- In terms of expected values, in general

$$\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$$

The left side $\mathrm{E}(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(\mathrm{E}(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 14.3. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Donny Dont says $\mathrm{E}(X) = \int_0^\infty (1/4.3)e^{-x/4.3} dx = 1$. Do you agree?

3. Compute and interpret $\mathrm{E}(X)$.

4. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

5. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

6. Donny Dont says: "I can just use LOTUS and replace x with x^2, so $\mathrm{E}(X^2)$ is $\int_{-\infty}^{\infty}(1/4.3)x^2e^{-x^2/4.3}dx$". Do you agree?

7. Compute $\mathrm{E}(X^2)$.

8. Compute $\mathrm{Var}(X)$.

9. Compute and interpret $\mathrm{SD}(X)$.

14.1 Exercises

Exercise 14.1. Continuous Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{E}(X)$.

2. Interpret $\mathrm{E}(X)$ in context.

3. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

4. Compute $\mathrm{Var}(X)$.

5. Compute and interpret $\mathrm{SD}(X)$.